



National University of Sciences and Technology (NUST)
School of Electrical Engineering and Computer Science

Faculty of Computing

CS-212: Object Oriented Programming

Class: BESE-16B

Assignment 1 – Semester Project Proposal

Date announced: 16th February 2026

Submission Deadline: 28th February 2026

Date of Proposal Presentation: **Week 7 (2nd March 2026)**

Instructor: Dr. M. Daud Abdullah Asif



Assignment 1: Project Proposal Submission

As part of your OOP course in Java programming, it is essential for you to undertake a semester project that will not only deepen your understanding of the programming language but also showcase your skills and creativity in applying Java concepts to solve real-world problems.

You are required to submit a project proposal outlining the following details:

1. **Title of the Project:** Provide a concise yet descriptive title for your project.
2. **Objective:** Clearly state the objective(s) of your project. What problem or challenge are you aiming to address or solve through this project?
3. **Scope:** Define the scope of your project. What functionalities and features do you plan to include? It's important to be realistic about what you can achieve within the given timeframe.
4. **Technologies and Tools:** Specify the technologies, frameworks, and tools you intend to use for developing your project. Since this is a Java project, ensure that Java will be the primary language used, but you may also mention additional libraries, databases, or APIs you plan to integrate.
5. **Project Deliverables:** List down the key deliverables of your project. What will you produce by the end of the project timeline (Final Project Submission: **Week 14** of this semester)?
6. **Team members (max 3):** Specify the team members by sharing a Google Sheet by this week and each should have a role and responsibilities throughout the project.

Please ensure that your proposal (**2-3 pages**) is well-written, organized, and addresses all the points mentioned above.

Some project Ideas:

Here are some of the sample project ideas that you might find interesting:

1. **Library Management System:** Develop a system to manage the operations of a library including book inventory management, user registration, borrowing and returning books, and generating reports.
2. **Online Shopping Platform:** Create a web-based application for online shopping where users can browse products, add them to cart, place orders, and make payments securely.
3. **Attendance Management System:** Build a system to automate the process of taking attendance in schools or organizations. It should allow teachers or administrators to mark attendance, view attendance records, and generate reports.
4. **Expense Tracker:** Design an application to help users track their expenses and manage their finances effectively. It should allow users to categorize expenses, set budgets, and generate financial reports.



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5. **Task Management Tool:** Develop a task management tool similar to Trello or Asana that allows users to create tasks, assign them to team members, set deadlines, and track the progress of tasks.
6. **Hospital Management System:** Create a comprehensive system for managing the operations of a hospital including patient registration, appointment scheduling, medical records management, and billing.
7. **Social Media Analytics Dashboard:** Build a dashboard that analyzes data from social media platforms such as Twitter or Facebook and provides insights into user engagement, sentiment analysis, and trending topics.
8. **Quiz or Exam Application:** Develop an application for conducting quizzes or exams online. It should allow teachers to create quizzes, assign them to students, and automatically grade the responses.
9. **Weather Forecasting Application:** Integrate with a weather API to create an application that provides weather forecasts for different locations. Users should be able to view current weather conditions as well as forecasts for the upcoming days.
10. **Recipe Management System:** Design a system for managing recipes where users can search for recipes, save their favorite recipes, create meal plans, and generate shopping lists based on selected recipes.

Feel free to **modify** any of these ideas or come up with your own project concept based on your interests and goals. You can also adapt new technologies such as Large Language models (LLMs) for creating chatbots or voice assistance by integrating their API but make sure that you create your own GUI in java.

Project Evaluation Criteria:

- Innovation and functionality (20%) – Novelty and relevance of the proposed project.
- User Interface (UI) Design (10%)
- Implementation and Code Quality (30%)
- Performance and Scalability (25%)
- Presentation & Documentation (15%) – Clarity, structure, adherence to the deadlines and professionalism. (including Proposal defense presentation)

Project Other Requirement:

- Ensure the number of students in your team is justifiable for the scope of the project.
- Each student should be responsible for a specific part of the project.
- Every student in the team should be able to answer questions about their designated part of the project.



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Deliverables

- Project Proposal Report (only one group member is allowed to upload the report via LMS)
- Deadline of submission is 28th February 2026, 11:59 PM
- Final Project Submission: 24th April 2026

Late submission will not be allowed since plenty of time is allocated for the project.
Submissions via email will not be marked